SOFTWARE REQUIREMENTS SPECIFICATION

Adaptive AI Assistant (JAS)

Normal JARVIS → Medical JARVIS → Mini JARVIS

Graduation Project

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Document structure based on IEEE Std 830-1998

1. Introduction

This document is the Software Requirements Specification (SRS) for the Adaptive AI Assistant graduation project. It defines the purpose, scope, users, functions, and requirements of the system across its three project layers: Normal JARVIS, Medical JARVIS (Alzheimer's Mode and ADHD Mode), the project showcase website, and the Mini JARVIS hardware prototype.

1.1 Purpose

This document defines the functional and non-functional requirements of the Adaptive AI Assistant. It is intended to give the project team, instructors, and reviewers a shared understanding of the system scope and expected behavior.

1.2 Scope

The software product being specified is the Adaptive AI Assistant, an AI assistant inspired by JARVIS. It is developed in three main layers:

- Normal JARVIS – a general voice-based assistant that interacts with the user, controls the operating system, opens/closes applications, sets timers, performs web searches, understands the current screen, and performs basic device actions.
- Medical JARVIS – a specialized extension of Normal JARVIS that adds Alzheimer's Mode and ADHD Mode, each changing JARVIS's voice, interface, interaction style, and available capabilities to match the needs of that user group.
- Mini JARVIS – a lightweight physical hardware prototype connected to the same JARVIS Core, demonstrating real hardware interaction without turning the project into a large hardware system.

The project also includes a lightweight project-showcase website. It is explicitly not a full web version of JARVIS.

The system will:

- Assist, remind, monitor, and support users through voice-first interaction across the described modes.
- Adapt its voice, interface, tone, and available features to the active mode (Normal, Alzheimer's, ADHD).

The system will NOT:

- Diagnose Alzheimer's disease, ADHD, or any medical condition.
- Provide medical treatment or replace professional medical care.
- Guarantee user safety; it assists and notifies but cannot ensure a hazard-free outcome.

[TBD] Exact target platforms (mobile/desktop/both), release scope for the graduation submission, and any commercial deployment plans.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
JARVIS	The Adaptive AI Assistant system, referenced by its working nickname (inspired by the fictional IRON MAN assistant).
JARVIS Core	The central processing system that powers all JARVIS variants (Normal, Medical, Mini).
Normal JARVIS	Phase 1 – the general-purpose voice assistant.

Term	Definition
Medical JARVIS	Phase 2 – the extension of Normal JARVIS with Alzheimer's Mode and ADHD Mode.
Alzheimer's Mode	Phase 2A – the specialized mode supporting users with Alzheimer's-related memory needs.
ADHD Mode	Phase 2B – the specialized mode supporting users with ADHD-related focus and organization needs.
Mini JARVIS	The lightweight physical hardware prototype connected to JARVIS Core.
Caregiver	An authorized person, such as a family member or caregiver, who may monitor the user's activity when this access is enabled.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
TBD	To Be Determined – a decision not yet finalized by the project team.

1.4 Organization

The remainder of this document is organized as follows: Section 2 gives an overall description of the product, its context, users, and constraints. Section 3 specifies the detailed functional and non-functional requirements, organized by project layer (Normal JARVIS, Alzheimer's Mode, ADHD Mode, Website, Mini JARVIS), followed by non-functional and cybersecurity requirements. Section 4 presents the modeling requirements, including use cases and representative scenarios. Section 5 describes the prototype. Section 6 lists references, and Section 7 gives the point of contact. Appendix A provides a traceability matrix, and Appendix B consolidates all open TBD items.

2. Overall Description

This section describes the product at a high level: its context, its main functions, its intended users, the constraints it operates under, and the assumptions and dependencies behind it.

2.1 Product Perspective

The Adaptive AI Assistant is a new, self-contained system rather than a component of an existing larger product. It consists of a central JARVIS Core (voice interaction, OS/device control, mode logic) accessed through: (a) a software assistant running on the user's device for Normal JARVIS, Alzheimer's Mode, and ADHD Mode; (b) a lightweight companion/showcase website; and (c) the Mini JARVIS hardware prototype, which connects to the same JARVIS Core over Wi-Fi.

Interface constraints:

- System interfaces: JARVIS Core exposes voice-processing and mode-switching logic to the software assistant and to Mini JARVIS.
- User interfaces: voice-first interaction for all modes; a simplified, low-complexity UI for Alzheimer's Mode; a minimal, low-distraction UI for ADHD Mode; a caregiver/doctor dashboard for monitoring.
- Hardware interfaces: Raspberry Pi, microphone, speaker, camera module, small display, LED/RGB indicator, and a physical emergency button on Mini JARVIS.
- Software interfaces: operating-system level APIs for app control and device actions; a web-search capability; [TBD] specific OS/platform and search provider.
- Communication interfaces: Wi-Fi connectivity between Mini JARVIS and JARVIS Core.

[TBD] Whether JARVIS Core is cloud-hosted, on-device, or hybrid; specific memory/storage and site-adaptation constraints.

2.2 Product Functions

At a high level, the Adaptive AI Assistant will:

- Provide natural, voice-based interaction with the user and understand the current screen (Normal JARVIS).
- Control the operating system, open/close applications, set timers, and perform web searches (Normal JARVIS).
- Support Alzheimer's users with memory, recognition, navigation, reminders, emotional support, and caregiver monitoring (Alzheimer's Mode).
- Support ADHD users with impulse control, prioritization, time management, focus support, and progress tracking (ADHD Mode).
- Present the project through a lightweight showcase website, including a lite interactive demo.
- Demonstrate real hardware interaction through the Mini JARVIS prototype.

Detailed functions are specified as functional requirements in Section 3.

2.3 User Classes and Characteristics

The following user roles interact with the system. Roles not needed by the current project scope (e.g., a general system administrator) are intentionally omitted.

Role	Description / Main Permissions
General User	Interacts with Normal JARVIS for everyday assistant functions (conversation, app control, timers, search, screen awareness). No specialized medical needs assumed.
Alzheimer's User	Uses Alzheimer's Mode. Benefits from a simplified UI, memory support, reminders, recognition, and emergency features. Assumed to need low-complexity interaction.
ADHD User	Uses ADHD Mode. Benefits from focus support, prioritization, pause-and-confirm safeguards, and progress tracking. Assumed to be capable of normal device interaction with attention-related support needs.
Caregiver	A family member or responsible person who monitors an Alzheimer's or ADHD user via the Caregiver/Doctor Dashboard and receives emergency alerts.
Doctor / Authorized Professional	Views monitored patterns and activity through the dashboard to support (not replace) professional judgment. Does not diagnose through the system itself.

2.4 Constraints

- The system is an assistive technology prototype; it must not present itself as a diagnostic or treatment tool for Alzheimer's, ADHD, or any medical condition.
- The system must not claim to guarantee user safety – emergency and monitoring features assist and notify but cannot ensure a hazard-free outcome.
- Alzheimer's Mode UI must remain low-complexity: large buttons, clear icons, few screens, simple navigation, consistent design.
- ADHD Mode UI must remain minimal and low-distraction: one primary action per screen, simple progress bar, reduced animation, voice-first interaction.
- The Mini JARVIS hardware prototype must remain lightweight and connect to the same JARVIS Core rather than becoming an independent large hardware system.
- The website must remain a lightweight project showcase, not a full web implementation of JARVIS.

2.5 Assumptions and Dependencies

- The user's device (or the Mini JARVIS Raspberry Pi) has a functioning microphone and speaker for voice interaction.
- An internet/Wi-Fi connection is available for web search, dashboard synchronization, and Mini JARVIS ↔ JARVIS Core communication.
- Family members' faces and voices used for recognition/personalization features are registered with informed consent.
- Caregivers/doctors have access credentials to the monitoring dashboard.

2.6 Deferred Requirements (Optional / Future Features)

The following are confirmed as optional/future-facing rather than core MVP scope, based on the Project File:

- ADHD Mode – Focus Levels (Light / Medium / Deep)
- ADHD Mode – Distraction Detection
- ADHD Mode – Focus Recovery

- ADHD Mode – Mood Check-in

All other listed Normal JARVIS, Alzheimer's Mode, ADHD Mode (core list), Website, and Mini JARVIS features are treated as core/MVP — see Section 3 priority labels and Appendix A.

3. Specific Requirements

This section enumerates the system's functional requirements by project layer, followed by non-functional requirements and cybersecurity requirements. Each functional requirement has a unique ID, a name, and a description that captures its trigger, expected system behavior, and expected output, plus a priority (Must / Should / Could Have). No optional feature listed in Section 2.6 has been elevated to Must Have.

3.1 Functional Requirements – Normal JARVIS (Phase 1)

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
FR-NJ-001	**Natural Voice Conversation** Trigger: user speaks to JARVIS. Behavior: system captures speech, interprets intent, and generates a natural spoken (and optionally text) response. Output: audible/visible reply relevant to the user's request.	Must Have
FR-NJ-002	**Open / Close Applications** Trigger: user issues a voice command naming an application. Behavior: system identifies the installed application and launches or terminates it. Output: application opens/closes; JARVIS confirms the action verbally.	Must Have
FR-NJ-003	**Basic OS Command Execution** Trigger: user requests a basic system action (e.g., volume, brightness, shutdown-type command). Behavior: system maps the request to a supported OS-level action and executes it. Output: action performed; confirmation given to user.	Must Have
FR-NJ-004	**Timer Creation & Management** Trigger: user asks JARVIS to set/cancel a timer by voice. Behavior: system creates, tracks, and alerts on the timer. Output: timer is set; an audible/visual alert fires at completion.	Must Have
FR-NJ-005	**Web Search** Trigger: user asks a question or requests a search. Behavior: system performs a web search and summarizes or reads back relevant results. Output: spoken and/or on-screen summary of search results.	Must Have
FR-NJ-006	**Screen Awareness & Explanation** Trigger: user asks JARVIS about what is currently on screen. Behavior: system reads/analyzes the current screen content and interprets it. Output: a spoken explanation describing the on-screen content.	Must Have
FR-NJ-007	**Installed Application Interaction** Trigger: user issues a command referencing a specific installed application's function. Behavior: system interacts with that application as supported. Output: requested in-app action performed or reported back.	Should Have
FR-NJ-008	**Device Function Control** Trigger: user requests control of an available device function (e.g., basic settings). Behavior: system executes the supported device-level function. Output: function applied; confirmation given.	Should Have

3.2 Functional Requirements – Alzheimer's Mode (Phase 2A)

Alzheimer's Mode is assistive: it supports, reminds, monitors, and notifies. It does not diagnose Alzheimer's disease or any related condition, and does not replace professional medical care.

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
FR-ALZ-001	**Question History & Pattern Tracking** Trigger: user asks JARVIS a question. Behavior: system logs the question with a timestamp, checks history for repetition, and answers naturally regardless of repeat count while tracking repetition frequency. Output: natural spoken answer; repetition pattern recorded for caregiver/doctor review.	Must Have
FR-ALZ-002	**Family Face Recognition** Trigger: a registered family member's face is detected by the camera. Behavior: system matches the face against stored family profiles. Output: JARVIS announces the recognized person's name to the user.	Must Have
FR-ALZ-003	**Item Location Memory** Trigger: user tells JARVIS where an item has been placed, or later asks where it is. Behavior: system stores the item-location association and retrieves the last known location on request. Output: spoken reminder of the last recorded location of the item.	Must Have
FR-ALZ-004	**Safe Navigation / Live Location Sharing** Trigger: user requests directions, or an the user requests navigation help or activates the emergency/location-sharing feature. Behavior: system provides navigation guidance and, in an emergency, shares the user's live location with configured contacts. Output: spoken directions; live location sent to caregiver contacts when triggered.	Must Have
FR-ALZ-005	**Persistent Date & Time Display** Trigger: main interface is active. Behavior: system continuously displays the current date, day, and time in a clear, always-visible format. Output: date/time shown persistently on the home screen.	Must Have
FR-ALZ-006	**Meal & Medication Reminders** Trigger: a scheduled meal or medication time is reached. Behavior: system issues a reminder including the medication name and image where applicable. Output: spoken/visual reminder delivered at the scheduled time.	Must Have
FR-ALZ-007	**Medication Confirmation** Trigger: a medication reminder has been delivered. Behavior: system asks the user to confirm the medication was taken and records the response. Output: confirmation status logged; unconfirmed doses are flagged for caregiver follow-up.	Must Have
FR-ALZ-008	**Emotional Support & Breathing / Pep Talk** Trigger: user expresses distress, sadness, or requests calming support. Behavior: system offers a short pep talk and/or guided breathing exercise. Output: calming spoken guidance delivered to the user.	Should Have
FR-ALZ-009	**Caregiver / Doctor Dashboard** Trigger: caregiver or doctor logs into the dashboard. Behavior: system presents activity, question-repetition patterns, medication adherence, and relevant alerts. Output: dashboard view of the user's monitored status over time.	Must Have
FR-ALZ-010	**Voice Personalization**	Should Have

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
	Trigger: caregiver/user configures a preferred familiar voice (e.g., a family member's voice). Behavior: system applies the selected voice profile to JARVIS's spoken output. Output: JARVIS speaks in the configured familiar voice.	
FR-ALZ-011	**Emergency Button / Voice Trigger** Trigger: user presses the emergency button or speaks a defined emergency phrase. Behavior: system immediately raises an emergency alert. Output: alert and location sent to configured emergency contacts.	Must Have
FR-ALZ-012	**Memory Book** Trigger: user opens the Memory Book or appears distressed/lonely. Behavior: system displays stored family photos, memories, and significant moments. Output: photos/memories shown or narrated to the user.	Should Have
FR-ALZ-013	**Alzheimer's-Optimized Interface** Trigger: Alzheimer's Mode is active. Behavior: system renders large buttons, clear icons, minimal screens, simple navigation, and a consistent layout. Output: simplified, low-complexity UI presented to the user.	Must Have

3.3 Functional Requirements – ADHD Mode (Phase 2B)

ADHD Mode is assistive: it supports focus, organization, and time management. It does not diagnose ADHD or treat any associated condition.

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
FR-ADHD-001	**Pause & Confirmation** Trigger: user issues a a command that matches a predefined action or rule selected by the system for confirmation. Behavior: JARVIS inserts a brief pause and asks for confirmation before executing. Output: action proceeds only after user confirms.	Must Have
FR-ADHD-002	**Smart Task Prioritization** Trigger: user adds tasks. Behavior: system orders tasks by priority, urgency, and deadline. Output: a prioritized task list presented to the user.	Must Have
FR-ADHD-003	**Smart Time Estimation & Time Alerts** Trigger: a task is started. Behavior: system estimates expected duration and monitors elapsed time. Output: alert sent to the user if a task runs longer than the estimate.	Must Have
FR-ADHD-004	**One Task at a Time** Trigger: user has multiple pending tasks. Behavior: system displays only the current task and advances to the next only once the current one is completed. Output: single-task focused view shown to the user.	Must Have
FR-ADHD-005	**Focus Mode** Trigger: user starts a focus session. Behavior: system reduces notifications and restricts access to distracting applications for the session duration. Output: distraction-reduced environment active until session ends.	Must Have
FR-ADHD-006	**Adaptive Voice Tone** Trigger: the user's reported mood/state or an explicit mood check-in. Behavior: system adjusts JARVIS's tone and speaking style accordingly. Output: voice responses delivered in a tone matched to the user's current state.	Should Have

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
FR-ADHD-007	**Stress Break & Breathing** Trigger: user reports or is detected to be under stress/overwhelm. Behavior: system reduces the current task's scope and suggests a break with a guided breathing exercise, then resumes the task. Output: guided break delivered; task resumed afterward.	Should Have
FR-ADHD-008	**Pep Talk** Trigger: during or after completion of a task/session. Behavior: system offers short encouragement. Output: motivational message delivered to the user.	Should Have
FR-ADHD-009	**Progress Bar** Trigger: user is working on a task or daily plan. Behavior: system tracks completion percentage. Output: visual progress bar showing current task/plan completion.	Must Have
FR-ADHD-010	**Medication Reminders** Trigger: scheduled medication time is reached. Behavior: system reminds the user and records whether the dose was confirmed as taken. Output: reminder delivered; adherence status logged.	Must Have
FR-ADHD-011	**User Dashboard** Trigger: user opens their dashboard. Behavior: system displays tasks, focus sessions, overall progress, and relevant activity. Output: consolidated activity/progress view shown to the user.	Must Have
FR-ADHD-012	**Focus Levels (Light / Medium / Deep)** Trigger: user selects a focus level before starting a session. Behavior: system applies the corresponding degree of restriction on notifications/apps. Output: focus session runs at the selected restriction level.	Could Have
FR-ADHD-013	**Distraction Detection** Trigger: user leaves the current task during an active focus session. Behavior: system detects the deviation and issues a gentle reminder to return. Output: reminder prompt guiding the user back to the task.	Could Have
FR-ADHD-014	**Focus Recovery** Trigger: user returns to a previously interrupted task. Behavior: system resumes from the last known progress point instead of restarting. Output: task resumed at its last saved state.	Could Have
FR-ADHD-015	**Mood Check-in** Trigger: a periodic check-in interval is reached. Behavior: system asks the user about their current state and uses the response to adjust interaction style. Output: mood recorded; interaction style adapted accordingly.	Could Have

3.4 Functional Requirements – Website

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
FR-WEB-001	**Home Page** Trigger: visitor opens the website. Behavior: system displays the project idea, the problem, and the proposed solution. Output: landing page with project introduction.	Must Have
FR-WEB-002	**Project Overview Page**	Must Have

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
	Trigger: visitor navigates to the overview section. Behavior: system presents a simplified explanation of the Adaptive AI Assistant, including summaries of Alzheimer's Mode and ADHD Mode. Output: overview content displayed.	
FR-WEB-003	**Features Page** Trigger: visitor navigates to the features section. Behavior: system lists the main capabilities of each mode (Normal, Alzheimer's, ADHD). Output: features page displayed.	Must Have
FR-WEB-004	**Team Page** Trigger: visitor navigates to the team section. Behavior: system displays team members and their roles. Output: team page displayed.	Must Have
FR-WEB-005	**Try JARVIS (Lite Demo)** Trigger: visitor opens the "Try JARVIS" section. Behavior: system provides a lightweight, in-browser demonstration of core JARVIS interaction. Output: interactive lite demo presented to the visitor.	Should Have

3.5 Functional Requirements – Mini JARVIS (Hardware)

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
FR-HW-001	**Voice Interaction (Raspberry Pi Core)** Trigger: user speaks near the Mini JARVIS device. Behavior: onboard microphone captures audio and forwards it to JARVIS Core for processing. Output: spoken response through the device speaker.	Must Have
FR-HW-002	**Timer on Mini JARVIS** Trigger: user requests a timer via the device. Behavior: device relays the request to JARVIS Core and displays timer status locally. Output: timer shown on the small display; audible alert at completion.	Should Have
FR-HW-003	**Face Recognition (Camera Module)** Trigger: a face is detected in the camera's field of view (Alzheimer's Mode). Behavior: device captures the image and forwards it to JARVIS Core for matching. Output: recognized family member's name announced and shown on the small display.	Should Have
FR-HW-004	**Emergency Button** Trigger: user presses the physical emergency button. Behavior: device immediately signals JARVIS Core to raise an emergency alert. Output: alert triggered without requiring voice input.	Must Have
FR-HW-005	**Focus Timer / Progress Display (ADHD Mode)** Trigger: an ADHD focus session is active. Behavior: device receives session/progress data from JARVIS Core. Output: focus timer and progress shown on the small display.	Should Have
FR-HW-006	**Status Indicator (LED / RGB)** Trigger: JARVIS Core changes state (listening, processing, speaking, alert). Behavior: device updates the LED/RGB indicator to reflect the current state. Output: visible status color/pattern on the device.	Must Have
FR-HW-007	**Small Display Output**	Must Have

ID	Requirement Name / Description (Trigger → Behavior → Output)	Priority
	Trigger: relevant status information is available (timer, recognized person, basic info). Behavior: device renders this information on its small display. Output: current status visible on-device.	
FR-HW-008	**Wi-Fi Connectivity to JARVIS Core** Trigger: device powers on or reconnects. Behavior: device establishes and maintains a Wi-Fi connection to JARVIS Core. Output: synchronized communication between Mini JARVIS and the rest of the system.	Must Have

3.6 Non-Functional Requirements

Exact numeric thresholds are not specified in the Project File and are marked [TBD] rather than invented; qualitative targets are given where reasonable.

Category	Requirement
Performance	JARVIS should respond to voice commands and reminders [TBD] Target response time for supported voice commands. for the user. [TBD] exact response-time targets (e.g., ms thresholds).
Usability	Alzheimer's Mode must be usable with minimal training (large buttons, simple navigation). ADHD Mode must minimize distraction and cognitive load per interaction.
Accessibility	Voice-first interaction should reduce reliance on fine motor input or complex reading. [TBD] compliance with a specific accessibility standard (e.g., WCAG level).
Reliability	Emergency Button / Voice Trigger and Medication Reminder features must function consistently, as they relate to user safety and health routines.
Security	Access to the Caregiver/Doctor Dashboard must be restricted to authorized caregivers/doctors. See Section 3.7 for detailed cybersecurity requirements.
Privacy	Face data, voice profiles, location data, and health-related logs (medication adherence, question patterns) must be stored and shared only with authorized parties (user, caregiver, doctor).
Maintainability	Requirements are organized by module (Normal JARVIS, Alzheimer's, ADHD, Website, Hardware) to allow each to evolve independently.
Compatibility	Mini JARVIS must communicate with JARVIS Core over Wi-Fi regardless of which host application/device runs the core assistant. [TBD] supported OS/device list.
Availability	Reminder, emergency, and dashboard features should remain available whenever the user's device/Mini JARVIS has connectivity. [TBD] target uptime.
Scalability	[TBD] – not specified in the Project File; relevant mainly if the dashboard is extended to multiple simultaneous caregiver/doctor users.

3.7 Cybersecurity Requirements

3.7.1 Threat Vectors

- Voice command spoofing or unauthorized voice input attempting to trigger OS/device actions.
- Unauthorized access to the Caregiver/Doctor Dashboard exposing health-related and location data.

- Interception of Wi-Fi communication between Mini JARVIS and JARVIS Core.
- Tampering with stored face-recognition profiles or medication/location data.
- Misuse of the Emergency Button/Voice Trigger to generate false alerts.

3.7.2 Threat Actors

- Unauthorized third parties without physical or network access rights to the device.
- Malicious actors on the same local network as Mini JARVIS.
- Unauthorized individuals attempting to view dashboard data without caregiver/doctor credentials.

3.7.3 Vulnerabilities & Mitigations

Vulnerability	Prevent / Detect / Mitigate
Weak or absent authentication on the Caregiver/Doctor Dashboard	Prevent: require authenticated login for dashboard access. [TBD] specific authentication mechanism.
Unencrypted Wi-Fi communication between Mini JARVIS and JARVIS Core	Prevent: use an encrypted communication channel. [TBD] specific protocol/standard.
Unauthorized emergency alert triggering	Detect/Mitigate: log all emergency triggers with timestamp and source for caregiver review.
Exposure of face, voice, or location data	Prevent: restrict access to authorized user/caregiver/doctor roles only; [TBD] encryption-at-rest approach.

4. Modeling Requirements

This section bridges the application domain (Section 3) and the machine domain through use cases and representative scenarios. Graphical diagrams (use case diagram, class diagram, sequence diagrams, state diagrams) are not yet produced and are marked [TBD]; the textual descriptions below define the content those diagrams will visualize.

4.1 Use Case Diagram

[TBD]

4.2 Use Case Descriptions

Each use case below refers to one or more functional requirements from Section 3.

ID	Use Case	Primary Actor(s)	Description	Related FR(s)
UC-01	Talk to JARVIS	General User	User speaks to JARVIS; system interprets intent and responds naturally.	FR-NJ-001
UC-02	Open an Application	General User	User asks JARVIS to open/close a named application.	FR-NJ-002
UC-03	Set a Timer	General User	User asks JARVIS to create a timer; JARVIS alerts on completion.	FR-NJ-004
UC-04	Perform Web Search	General User	User asks a question; JARVIS searches the web and reports results.	FR-NJ-005
UC-05	Understand Current Screen	General User	User asks about on-screen content; JARVIS analyzes and explains it.	FR-NJ-006
UC-06	Handle Repeated Question	Alzheimer's User / Caregiver	User re-asks a previously asked question; JARVIS answers naturally and logs the repetition.	FR-ALZ-001
UC-07	Recognize Family Member	Alzheimer's User	Camera detects a registered face; JARVIS announces who it is.	FR-ALZ-002
UC-08	Find a Stored Item	Alzheimer's User	User asks where an item is; JARVIS recalls the last recorded location.	FR-ALZ-003
UC-09	Medication Reminder	Alzheimer's User / ADHD User	Scheduled time is reached; JARVIS reminds the user and records confirmation.	FR-ALZ-006/007, FR-ADHD-010
UC-10	Emergency Alert	Alzheimer's User / Caregiver	User presses the emergency button or speaks the trigger phrase; alert and location are sent to contacts.	FR-ALZ-004/011, FR-HW-004

ID	Use Case	Primary Actor(s)	Description	Related FR(s)
UC-11	Use Memory Book	Alzheimer's User	User opens the Memory Book to view family photos and memories.	FR-ALZ-012
UC-12	Prioritize ADHD Tasks	ADHD User	User adds tasks; JARVIS orders them by priority, urgency, and deadline.	FR-ADHD-002
UC-13	Start Focus Mode	ADHD User	User starts a focus session; distractions are reduced until it ends.	FR-ADHD-005
UC-14	Pause & Confirmation	ADHD User	User issues a potentially impulsive command; JARVIS pauses and asks for confirmation.	FR-ADHD-001
UC-15	Distraction Detection	ADHD User	User drifts from the current task during focus mode; JARVIS gently redirects them.	FR-ADHD-013
UC-16	Focus Recovery	ADHD User	User returns to an interrupted task; JARVIS resumes from the last saved progress.	FR-ADHD-014
UC-17	Caregiver Monitoring	Caregiver / Doctor	Caregiver/doctor opens the dashboard to review activity, patterns, and alerts.	FR-ALZ-009, FR-ADHD-011
UC-18	Mini JARVIS Interaction	General / Alzheimer's / ADHD User	User interacts with the physical Mini JARVIS prototype (voice, camera, button, display).	FR-HW-001 to FR-HW-008

4.3 Class Diagram

[TBD]

4.4 Representative Scenarios

Scenario: Handle Repeated Question (UC-06)

An Alzheimer's User asks JARVIS, “What time is my appointment?” for the third time in an hour. JARVIS answers naturally each time, without indicating irritation or repetition to the user, while logging the question and timestamp internally. The caregiver later reviews the repetition pattern on the Caregiver Dashboard.

Scenario: Start Focus Mode with Pause & Confirmation (UC-13, UC-14)

An ADHD User starts a Focus Mode session on a task. Mid-session, the user issues a command to open an unrelated, potentially distracting application. JARVIS pauses and asks the user to confirm before proceeding, reducing impulsive task-switching. The Progress Bar continues to reflect the original task's completion status.

Scenario: Emergency Alert via Mini JARVIS (UC-10, UC-18)

An Alzheimer's User presses the physical Emergency Button on the Mini JARVIS device. The device signals JARVIS Core, which immediately sends an alert with the user's live location to the configured emergency contacts and caregiver, and updates the LED indicator to an alert state.

4.5 State Diagrams

[TBD]

5. Prototype

[TBD]

5.1 How to Run Prototype

[TBD]

5.2 Sample Scenarios

Mini JARVIS Hardware Prototype

The project includes a compact Mini JARVIS device designed as a lightweight physical interface for the assistant. The prototype combines voice interaction, a camera, display, and speaker in one device. It is intended to support selected features from the Normal JARVIS and medical modes.



Figure 5.1 — Mini JARVIS hardware prototype and proposed system architecture.

6. References

- **[1]** IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications (SRS template basis for this document).
- **[2]** Mayo Clinic, *Adult ADHD: Symptoms and Causes*. (Reference for ADHD-related symptoms including impulsivity, difficulty with attention, prioritization, time management, multitasking, planning, self-control, mood fluctuations, task completion, and coping with stressful situations.)
- **[3]** American Psychiatric Association, *What is ADHD?* (Reference for the general definition, symptoms, and characteristics of Attention-Deficit/Hyperactivity Disorder.)
- **[4]** Alzheimer's Association, *What is Alzheimer's Disease?* (Reference for Alzheimer's disease, its effects on memory, thinking, behavior, and daily functioning.)